

CodeWeaver comparison with CRUST-Bench

On all 100 exact CRUST-Bench subjects, 300/300 CodeWeaver cells compiled and 165/300 passed every fixed project test. Mean project success was 55.00% (sample SD 1.00 pp). The paper's single-shot and three-round repair settings are preserved as references; CodeWeaver's multi-stage five-repair/three-parity protocol is not relabeled as those settings.

Method

This exact-subject re-analysis imports the published CodeWeaver campaign's independently restored CRUST interfaces and fixed tests. All three terminal outcomes are retained; no best-of-three selection is used.

Validity boundary

CRUST-Bench's Table 4 reports pass rates under single-shot, compiler repair, test repair, and an adapted SWE-agent. CodeWeaver uses a different architecture and larger repair budget. Comparison is descriptive, not a controlled model ablation.

Exact CodeWeaver measurements

Run	Build	Pass all	Fixed tests	Test rate
CodeWeaver rep 1	100/100	56/100 (56.00%)	447/623	71.75%
CodeWeaver rep 2	100/100	55/100 (55.00%)	454/623	72.87%
CodeWeaver rep 3	100/100	54/100 (54.00%)	423/623	67.90%
CodeWeaver mean	100.00%	55.00% +/- 1.00 pp	three independent repetitions	95% t-CI +/- 2.48 pp

Published CRUST-Bench Table 4

System	Base build	Base test	Compiler-repair build	Compiler-repair test	Test-repair build	Test-repair test
OpenAI o3	35.00%	19.00%	68.00%	31.00%	63.00%	48.00%
Claude Opus 4	43.00%	22.00%	78.00%	29.00%	65.00%	40.00%
OpenAI o1	32.00%	15.00%	69.00%	28.00%	54.00%	37.00%
Claude 3.7	26.00%	13.00%	54.00%	23.00%	49.00%	32.00%
Claude 3.5	26.00%	11.00%	49.00%	21.00%	38.00%	24.00%
o1-mini	19.00%	9.00%	47.00%	16.00%	27.00%	21.00%
GPT-4o	18.00%	7.00%	52.00%	18.00%	42.00%	22.00%
Gemini 1.5 Pro	11.00%	3.00%	35.00%	11.00%	30.00%	14.00%
Virtuoso (Distilled Deepseek V3)	2.00%	2.00%	21.00%	6.00%	10.00%	6.00%
Deepseek-Coder-32B	1.00%	0.00%	2.00%	0.00%	2.00%	0.00%

System	Base build	Base test	Compiler-repair build	Compiler-repair test	Test-repair build	Test-repair test
QwQ-32B-Preview	1.00%	0.00%	1.00%	0.00%	1.00%	0.00%
Qwen-2.5-Coder-32B	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Adapted SWE-agent (Claude-3.7)	41.00%	32.00%	N/A	N/A	N/A	N/A

CodeWeaver execution and model-use telemetry

Scope	Cells	Elapsed h	Assistant turns	Tool calls	Premium requests	AIU	Output tokens	Input tokens
repetition 1	100	157.28	47620	75442	1438	225874.65	27560480	N/A (unavailable)
repetition 2	100	168.33	50605	80319	1503	240674.81	29239390	N/A (unavailable)
repetition 3	100	143.57	47989	76484	1390	229364.16	27616255	N/A (unavailable)
all measured cells	300	469.18	146214	232245	4331	695913.62	84416125	N/A (unavailable)

CodeWeaver coverage measurements

Rep	Metric	Cells	Mean	Min	Max
1	coverage_before	96	56.86%	0.00%	100.00%
1	coverage_after	96	57.81%	0.00%	100.00%
2	coverage_before	96	59.22%	0.00%	100.00%
2	coverage_after	96	59.67%	0.00%	100.00%
3	coverage_before	96	57.25%	0.00%	100.00%
3	coverage_after	96	58.28%	4.58%	100.00%

CodeWeaver final-output Clippy measurements

Rep	Complete	Incomplete	Warning-free	Warnings	Errors	Alerts/project	Alerts/function
1	96	4	1	1959	0	20.41	0.183
2	95	5	1	1827	0	19.23	0.166
3	92	8	0	2002	0	21.76	0.187

Complete source-paper surface audit

Surface	Denominator	Metrics	Artifact status
Table 1	7 benchmarks	projects, multi-file, average LoC, Rust interface, Rust tests	structured_reference
Figure 2	100 projects	seven application categories	structured_reference
Table 2	100 projects	tests, files, coverage, LoC, pointer dereferences, functions	structured_reference
Table 3	299 files / 3,085 functions	files, functions, arguments, ownership/type features	structured_reference
Table 4	100 projects	build and test success	structured_reference_and_c odeweaver_comparison
Table 5	percent of 100 projects	mismatch, borrow, missing, unimplemented, trait, arguments, unsafe	structured_reference
Appendix A	100 projects	representativeness and limitations	documented_reference
Appendix B	10-project pilot; six model calls per task	qualitative winner; numeric cells not reported	documented_reference
Appendix E Table 6	adapted SWE-agent experiment	build/test by step budget	paper_only